

Mental Models

What is it?

People develop internal expectations about how things should work based on prior experience. These expectations influence how they interact with new products or systems.

History

Discussed across cognitive psychology and HCI research since around **1983**, showing that users transfer familiar patterns from past environments into new interfaces to reduce cognitive effort.

The Psychology Behind It

- People reuse previously learned solutions
- Familiar structures reduce thinking
- Misaligned expectations create confusion
- Mental models drive how users interpret functions, layouts, and flows

Why It Matters

- Risks inaccurate design decisions
- Misalignment reduces usability
- Cultural differences influence assumptions
- Wrong models lower quality engagement

How to Apply It

- Understand user expectations via surveys, interviews, journey mapping
- Align with familiar established patterns
- Strengthen mental models with clear labels, analogies, and metaphors
- Educate gently through microcopy, tutorials, progressive learning

Theory in Action

- Amazon aligns checkout flows with universally expected shopping patterns
- Cars use consistent ignition actions (key turn or push button) across all brands to match existing mental models

Final Thought

Great design fits the way users already think.

When you design toward mental models instead of against them, everything feels natural and effortless.